

**BETTERBLINDS: AN AUTOMATED SUNLIGHT-RESPONSIVE BLINDS
FOR ENHANCING CLASSROOM LEARNING
ENVIRONMENT**

A Research
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by

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APPROVAL SHEET

In fulfillment of the requirements for the 3I - Work Immersion of Science Technology, Engineering and Mathematics strand, this research entitled **“BETTERBLINDS: AN AUTOMATED SUNLIGHT-RESPONSIVE BLINDS FOR ENHANCING CLASSROOM LEARNING ENVIRONMENT”** prepared and submitted by Julz Clar Leoniel E. Alvarado, Marco Sebastian S. Casinillo, Arth Andrey C. Endrina, and Vanessa Angelica Sieras, has been examined, accepted, and approved for PROPOSAL HEARING.

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ABSTRACT

Many students are affected by direct sunlight indoors due to improper shading, impacting their performance. Research indicates widespread issues with improper shading in schools and workplaces. Traditional manual blinds are inefficient in responding to changing sunlight conditions, leading to poor learning experiences. This research addresses the need for an automated solution to enhance classroom learning environments. This study aims to evaluate the feasibility and effectiveness of implementing “BetterBlinds: An Automated Sunlight-Responsive Blinds for Enhancing Classroom Learning Environment” in educational settings. In this context, classroom learning environment is defined as the physical, social, and emotional conditions within a classroom, including factors like layout, lighting, temperature, noise levels, seating arrangements, and available resources. To test the hypothesis that classrooms equipped with "BetterBlinds" will demonstrate significantly improved learning outcomes compared to classrooms without the blinds, the study involved designing and prototyping BetterBlinds, integrating LDRs, motors, and control systems. Then, classroom trials were conducted to assess the system's effectiveness in reducing glare and maintaining comfortable lighting. Initial findings indicate that BetterBlinds efficiently adapts to sunlight changes, reducing glare and ensuring optimal classroom lighting. Results indicate that BetterBlinds effectively reduces sunlight glare, enhancing visibility and creating a more favorable learning environment. This innovation represents a promising solution for optimizing classroom conditions and facilitating enhanced learning outcomes. Recommendations include further research on user feedback and potential adaptations for broader educational settings.

Keywords: BetterBlinds, Automated blinds, sunlight responsive, classroom learning environment, lighting optimization, learning enhancement

INTRODUCTION

A comfortable environment is crucial in academics as it helps students, teachers, and other office staff think properly and efficiently. During this time, many students and office workers are currently suffering from direct sunlight through windows indoors due to improper shading, which could alter the course of how they work and study. This event creates distraction and disturbances, such as humidity and direct exposure to sunlight which can be irritating and disturbing.

Proper shading plays an important role in maintaining the flow of comfort in an environment. Research has shown that many schools or working establishments have been dealing with improper internal or external shading. However, little research on visual comfort relating to the potential effects of it on the student's overall performance has been done in educational buildings globally as compared to other types of buildings such as working places. Hence, this study addresses this gap and aims to offer valuable findings about the effectiveness and feasibility of implementing automated sunlight-responsive blinds in enhancing the classroom learning environment in educational settings.

In summary, a comfortable academic environment is crucial for productivity, but improper shading, especially from direct sunlight, creates distractions and disturbances. Studies highlight the importance of sun control devices like blinds in managing heat gain. Shading also contributes to visual comfort addressing issues such as over-illumination. Research indicates widespread challenges in schools and workplaces, emphasizing the impact of daylight on student performance. While the focus is providing proper shading, the need for a balanced approach in designing shading devices is stressed, considering factors like building type and cost-effectiveness. In essence, proper shading plays a pivotal role in achieving optimal comfort and productivity in educational settings.

Theoretical Background

This study is anchored on three theories, namely; the Photoconductivity Theory, Lorentz Force Theory, and Light Control Theory.

Photoconductivity Theory

The first theory, Photoconductivity Theory (PT), examines the increase in the electrical conductivity of a material when it absorbs photons (light particles) with sufficient energy. In the case of a photoresistor, it consists of a semiconductor material that changes whenever light falls on it. This variance in conductivity is utilized to detect light levels, which can be translated into control signals for the blinds. Gudden and Pohl (1920) developed the PT. The study's product relies on the understanding and application of PT to function effectively. A photoresistor is important so that the study's blinds system can automatically respond to changing light conditions.

Lorentz Force Theory

Another theory is the Lorentz Force Theory (LFT). Lorentz Force Theory is a law that states that a conductor carrying a current, like the coils in the motor's rotating part (armature), experiences a force when placed in a magnetic field (generated by the motor's stationary part, stator). This is the theory that is behind the operation of DC Motor. This operates based on the interaction between magnetic fields and conductors that are carrying current, following Faraday's law of electromagnetic induction. The direction of rotation is determined by the polarity of the applied voltage and the orientation of the magnetic field. Thus, by varying the magnitude and polarity of the applied voltage, the speed and direction of rotation of the DC motor can be controlled. In the context of the study, DC motors serve as the powerhouse in automated blinds systems, translating electrical energy into mechanical motion to achieve precise and responsive control of blind positioning. By understanding the principles of DC motor

operation and optimizing their integration within the system, the researchers can create automated blinds solutions that enhance user comfort, energy efficiency, and overall functionality.

Light Control Theory

The last theory is the Light Control Theory (LCT). This theory suggests that blinds, typically Venetian blinds, are designed to regulate the amount of light entering a room by adjusting the angle of the “slats”, a thin, narrow piece of wood, plastic, or metal used to make floors, furniture, window coverings, etc. This functionality is based on the principle of light control, which involves manipulating the orientation of the slats to either allow light to pass through or block it entirely. By adjusting the angle of the slats, users can control the direction and intensity of incoming light, thereby creating desired lighting conditions and enhancing comfort within the room. The understanding of the functionality of these types of blinds provides a foundation for addressing key considerations, one of which is the light control, in the design and implementation of the study’s blinds systems.

Review of Related Literature

This chapter covers a review of related literature and studies chosen by the researchers that are particularly beneficial throughout the studies.

Numerous studies have shown the significance of daylight in buildings. Daylight may have positive impacts on view, visual, comfort, psychological comfort, health, and workplace productivity (Boyce, P. et al., 2003). However, once uncontrolled, it can have drastic negative impacts on the building environment. With shading devices, they provide the opportunity to regulate daylight, maximizing its control and benefits of daylight penetrating into buildings. In particular, automated blinds have been used recently to overcome the limitations of manual operation.

In the (2010) paper of Koo, S.Y. et al., they provided a new method for automated venetian blinds control that aims to maximize daylight penetration while also preventing direct solar glare. Results of the study have shown that there are more potential benefits on the new control method compared to the previous method with respect to energy savings and occupant comfort.

On the contrary, a study by O'Brien, W. et al. (2013) provides a comprehensive and critical review of experimental and study methodologies for manual shade operation in office buildings, their results, and their application to building design and controls. Furthermore, the study reviewed four major topics related to occupant use of manually controlled shades in office buildings, namely: observational study methodologies, empirical results, applications of the results, and finally, façade design principles. Among all, twelve main studies were reviewed and generally drew the same qualitative conclusions, with some minor exceptions. Major conclusions of the study have shown that the biggest motivation for closing shades is to prevent glare and other forms of visual discomfort. Also, if occupants have control over their lighting or if their lights are automated to turn off during periods of adequate daylight, they are more inclined to control shades to achieve desirable lighting conditions.

METHODOLOGY

The research was conducted through experimentation, consisting of three distinct categories of trials, each comprising five individual trials. The testing was conducted at the residence of one of the researchers. It is located in Blk 9 Lot 4, La Aldea Buena Mactan, Timpolok, Lapu-Lapu City.

Table I. Cost of Materials

Equipment/Material	Quantity	Cost
Blinds	1	₱ 650.00 (Price may vary depending on the scale and shape)
Arduino Board	1	₱ 450.00
DC Motor	2	₱ 520.00
LDR Sensor Module	1	₱ 110.00
Connecting/Jumper Wires	30	₱ 150.00
Resistor	5	₱ 75.00
USB Power Adapter & USB to DC Power Cable	1	₱ 280.00
Breadboard PCB	1	₱ 76.00
L293D Motor Driver Shield	1	₱ 129.00
5V Relay Module	1	₱ 42.00
Capacitor 16V 100uF	1	₱ 80.00
Soldering Lead	1	₱ 55.00
		Total: ₱2,617.00

Product Design

Figure I. Internal Design

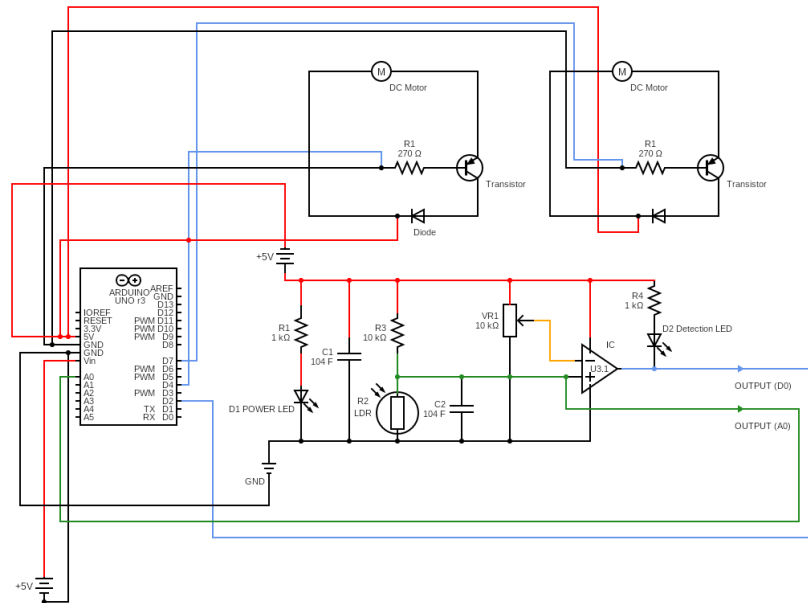
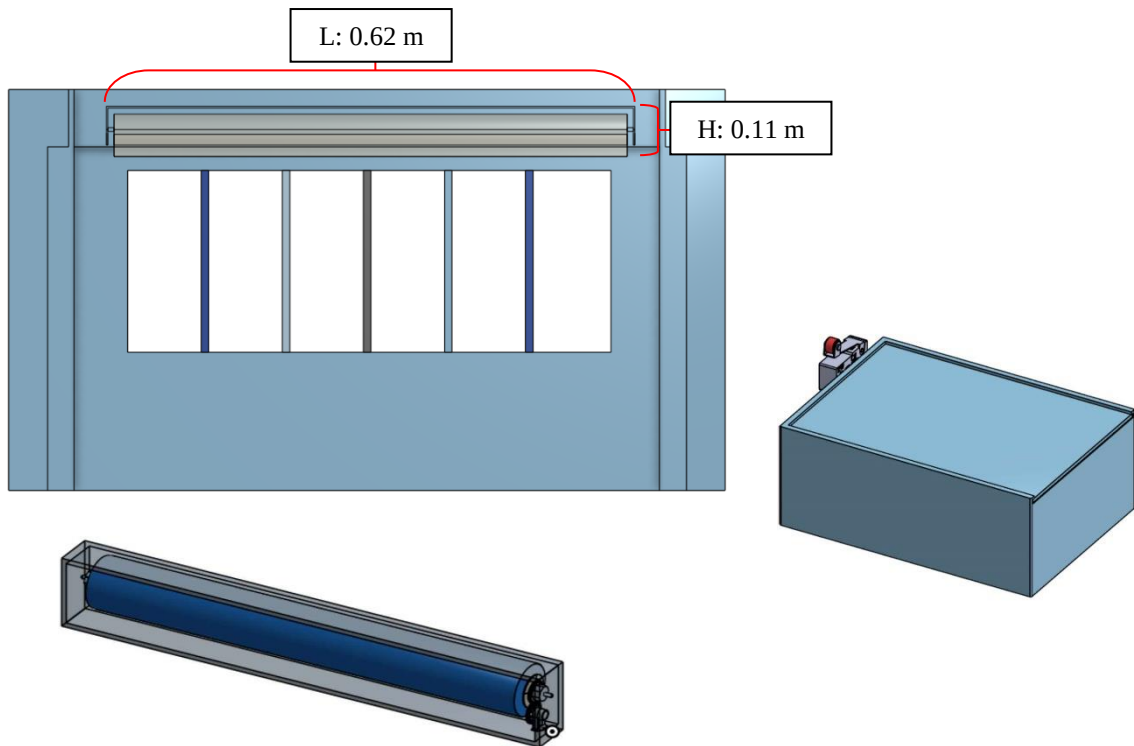


Figure II. External Design



Research Procedure

Table II. Research Procedures

STEPS	PROCEDURE
STEP 1	Purchased all the necessary materials and components listed below from local & online stores.
STEP 2	Attached the DC motor to the blinds mechanism, ensuring proper alignment.
STEP 3	Mounted the photoresistor in a strategic position to detect sunlight.
STEP 4	Then, the control box was placed in a convenient and discreet location.
STEP 5	The Arduino board was connected to the DC motor using the L293D Motor Driver Shield.
STEP 6	Connected the photoresistor to the Arduino board for light detection.
STEP 7	The 5V relay module was set up to control the blinds' on/off function.
STEP 8	Installed the capacitor to stabilize voltage fluctuations in the circuit.
STEP 9	The Arduino code is then written and uploaded to control the blinds based on light levels.
STEP 10	Programmed the Arduino to read data from the photoresistor and actuate the motor.
STEP 11	Lastly, calibrations were made in the code to adjust the sensitivity of the light detection.

Experimental Design

Three distinct sets of trials were conducted to evaluate the product's performance in various aspects, including the precision of the photoresistor's threshold, the speed of the DC Motor's rotation, and the accuracy of the DC Motor in positioning the blinds correctly. The experimentation involved fine-tuning the photoresistor's sensitivity to light to establish the threshold for activating the blinds' movement. Additionally, multiple trials were conducted to determine the optimal speed of the DC Motor, ensuring it does not cause damage to the product's rolling mechanism. In total, five trials were carried out for each category of experimentation.

Trials conducted inside the classroom revealed that the photoresistor registered values ranging from 700 to 900 under harsh light conditions, with an average value of 819.2 across five trials. Conversely, normal light conditions yielded values between 200 and 300, averaging 249.8 from the same number of trials. Based on these findings, it was determined that the threshold for activating the DC Motor to roll down the blinds should be set at 819 or above, while the threshold for rolling up the blinds should be 250 or below. For the DC Motor's speed, during the initial three trials, it was evident that the DC motor rotated at excessively high speeds, posing a risk of damaging the blinds due to rapid movement. Because of this, it was determined that the ideal speed for the DC motor should strike a balance between not being too fast and not being too slow. While the fourth trial showed a slight reduction in speed, it remained too fast for optimal operation. However, the fifth trial yielded the desired outcome, as the DC motor operated at a speed that effectively rolled the blinds without causing damage. In the final phase of trial testing, across five trials, it was found that the optimal delay time for rolling down the blinds to completely block out light was determined to be 4,170 milliseconds, or 4.17 seconds. Similarly, for rolling up the blinds to their starting position, the ideal delay time was found to be 4,890 milliseconds, or 4.89 seconds. These findings were consistent across all trials, indicating the precise timing required for the blinds to achieve their desired positions effectively.

RESULTS AND DISCUSSIONS

Table VI. Photoresistor Ideal Threshold

Trial No.	Photoresistor Value Under Harsh Light Conditions (Blinds will roll down)	Photoresistor Value Under Normal Light Conditions (Blinds will roll up)
1	878	258
2	702	234
3	759	263
4	865	253
5	892	241

Table VII. DC Motor's Speed

Trial No.	DC Motor's Rotation Speed	Assumptions
1	Excessively Fast	Can potentially damage the product
2	Excessively Fast	Can potentially damage the product
3	Excessively Fast	Can potentially damage the product
4	Moderately Fast	Minimal damage but still pose risks
5	Moderately Right	Suitable

Table VIII. Optimal Delay Time for Rolling Down

Trial No.	Delay Time	Outcomes
1	4,450 ms / 4.45 s	The blinds rolled down way beyond and exceeded the ideal spot for blocking out light.
2	4,230 ms / 4.23 s	The blinds rolled down way beyond and exceeded the ideal spot for blocking out light.
3	4,050 ms / 4.05 s	The blinds didn't completely roll down to effectively block out light.
4	4,170 ms / 4.17 s	The blinds completely rolled down in a manner that just perfectly blocks out light.
5	4,150 ms / 4.15 s	The blinds completely rolled down in a manner that just perfectly blocks out light.

Table IX. Optimal Delay Time for Rolling Up

Trial No.	Delay Time	Outcomes
1	5,200 ms / 5.2 s	The blinds exceedingly retracted way beyond their initial position.
2	4,150 ms / 4.15 s	The blinds didn't fully retract back to their initial position.
3	4,460 ms / 4.46 s	The blinds didn't fully retract back to their initial position.
4	4,750 ms / 4.75 s	The blinds didn't fully retract back to their initial position.
5	4,890 ms / 4.89 s	The blinds fully retracted and returned to their initial position.

This system is based on Arduino UNO and a photoresistor. The system automatically triggers the motor to rotate counterclockwise when the reading value of the photoresistor surpasses the set threshold for harsh light conditions, and vice versa when it falls below the set threshold for normal light conditions.

The direction in which the motor will rotate is controlled by the Motor Driver Shield component. The system is programmed to rotate the motor counterclockwise, lowering the blinds, when the photoresistor detects a value surpassing the threshold for harsh light conditions, indicating the presence of solar glare. Conversely, it rotates clockwise to raise the blinds when the photoresistor detects a value equal to or below the threshold for normal light conditions, signaling the absence of solar glare.

Based on the trials conducted, the average value detected by the Photoresistor under harsh light conditions is 819.2, while under normal light conditions, it is 249.8. Consequently, these values were used as the optimal thresholds to trigger the rotation of the DC motor and determine its direction based on the prevailing light conditions.

Additionally, it is crucial to consider the total operation time of the blinds to ensure the accuracy and smooth functionality of the system. Through testing, it was determined that it takes a delay of 4,170 milliseconds or 4.17 seconds for the blinds to fully roll down and effectively block out light. Conversely, it takes a delay of 4,890 milliseconds or 4.89 seconds for the blinds to retract fully and return to their initial position. Further trials were conducted using these values to confirm their accuracy, and consistent results were observed throughout. The blinds didn't fully retract back to its initial position.

CONCLUSION

In summary, the product effectively provides shade with minimal effort. When the Photoresistor detects light and reaches the programmed luminosity threshold, it sends a signal to the Arduino, prompting the DC motors to rotate the blinds accordingly. Once the blinds have been lowered, they automatically retract to their original position after a certain period. This occurs when the Photoresistor no longer detects light or when the luminosity level drops below a specific threshold programmed into the Arduino. By effectively regulating the amount of sunlight entering homes, these blinds effectively control indoor temperatures and reduce glare, creating a more pleasant environment for daily activities. Residents no longer need to manually adjust blinds throughout the day, as the automated system seamlessly responds to changing light conditions, providing optimal shade and comfort with minimal effort. This accessibility and convenience are especially beneficial for individuals with mobility limitations or busy lifestyles, offering a hassle-free solution to shade control.

RECOMMENDATIONS

- 1) Implementing remote control system through signal input.
- 2) It is recommended to use Korean blinds since it can easily roll with the DC motor due to its design being double sided.
- 3) It is recommended to use DC motors that are compatible with the type of blinds that are currently used, ensuring smooth and efficient operation.
- 4) Incorporate scheduling features into the remote-control system, allowing users to program specific times for opening, closing, or adjusting the blinds automatically. This enhances convenience and energy efficiency.

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APPENDIX E - BIONOTE

JULZ CLAR LEONIEL E. ALVARADO



Julz Clar Leoniel E. Alvarado, 2022-2023, was born on April 29, 2006 at Cebu City, Cebu. He is a senior high school student at University of Cebu Lapu-Lapu and Mandaue. He is a consistent honor student during his elementary years at San Fernando Elementary School and Babag 1 Elementary School. During high school he participated in School Press Conference in the Editorial Cartooning English category and won the School, District, Division levels. Julz intends to take an examination for a scholarship. He plans to pursue Bachelor of Science in Architecture to become a architect.

MARCO SEBASTIAN S. CASINILLO



Marco Sebastian S. Casinillo is a Grade 12 student at the University of Cebu Lapu-Lapu and Mandaue. With a passion for exploring new ideas and concepts, he thrives in the academic environment. His elementary years were marked by notable achievements with honors, showcasing his dedication and hard work. Transitioning into junior high school, he maintained an average performance. Despite the challenges, he remains committed to his education and looks forward to exploring new opportunities in the future.

ARTH ANDREY C. ENDRINA



Arth Andrey C. Endrina is a dedicated Grade 12 student at University of Cebu Lapu-Lapu Mandaue, demonstrating consistent academic excellence throughout his education. Notably, he achieved honors throughout elementary and high school, while also securing accolades as a placer in district and division presscon copyreading. His diverse talents extend to winning the school-level Sudoku competition, showcasing his aptitude beyond the academic realm.

VANESSA ANGELICA SIERAS



Vanessa Angelica Sieras is an 18-year-old Grade 12 STEM student at the University of Cebu Lapu-lapu and Mandaue. With a consistent track record of academic excellence, she has been recognized as an honor student since 2018. As the former Supreme Student Government President, Vanessa received the Service Award for her dedication during the 2021-2022 term. Additionally, she was honored with the JCI-Youth Leadership Excellence Award in 2022 and served as a Microsoft Youth Ambassador, advocating for technology literacy. Vanessa's educational journey began at Paknaan National High School and Tultugan Elementary School, where she nurtured her commitment to academic success.